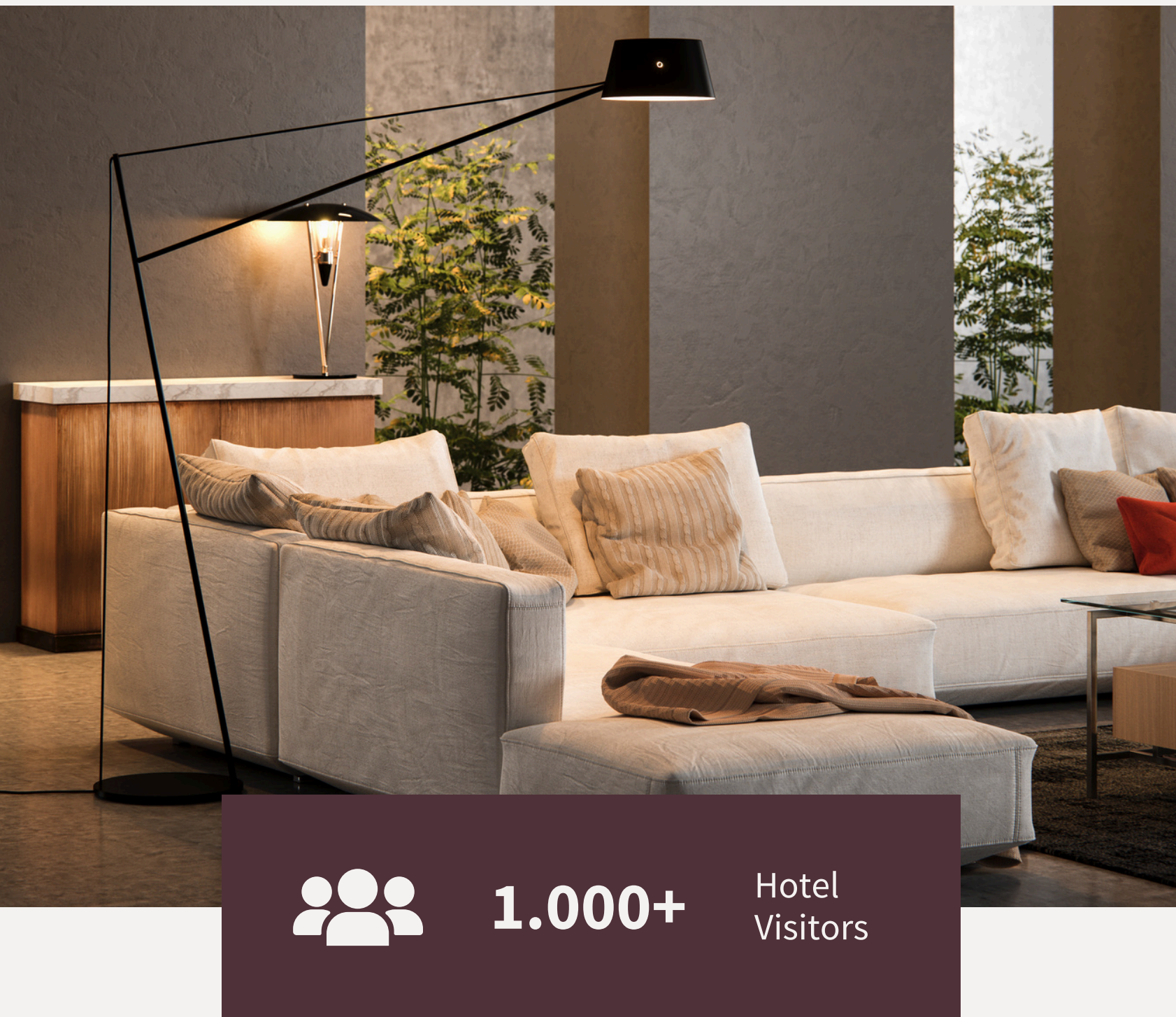




HOTEL

MANAGEMENT SYSTEM



Purpose of the System

Hotel Management System

The main objective of the Hotel Management System (HMS) is streamlining basic hotel functions specifically room booking, check-in and checkout of guests, billing and various types of reporting. As manual tasks are minimized, mass accommodation processes are improved resulting in higher levels of guest satisfaction.



Scope & System Overview

Scope

This document focuses on hotel room booking, customer management, billing, inventory and reports. Note that such third- party integrations are out of scope (e.g. travel agencies, gateways to third parties e.g. payments platforms)

System Overview

The HMS, according to its original concept, is an autonomous system that performs the core functions of a hotel business such as coordinating room reservations, checking guests in and out, handling financial management, and issuing reports. While this system helps to provide a seamless customer experience by interlinking with internal hotel systems such as the inventory system and the accounting system.

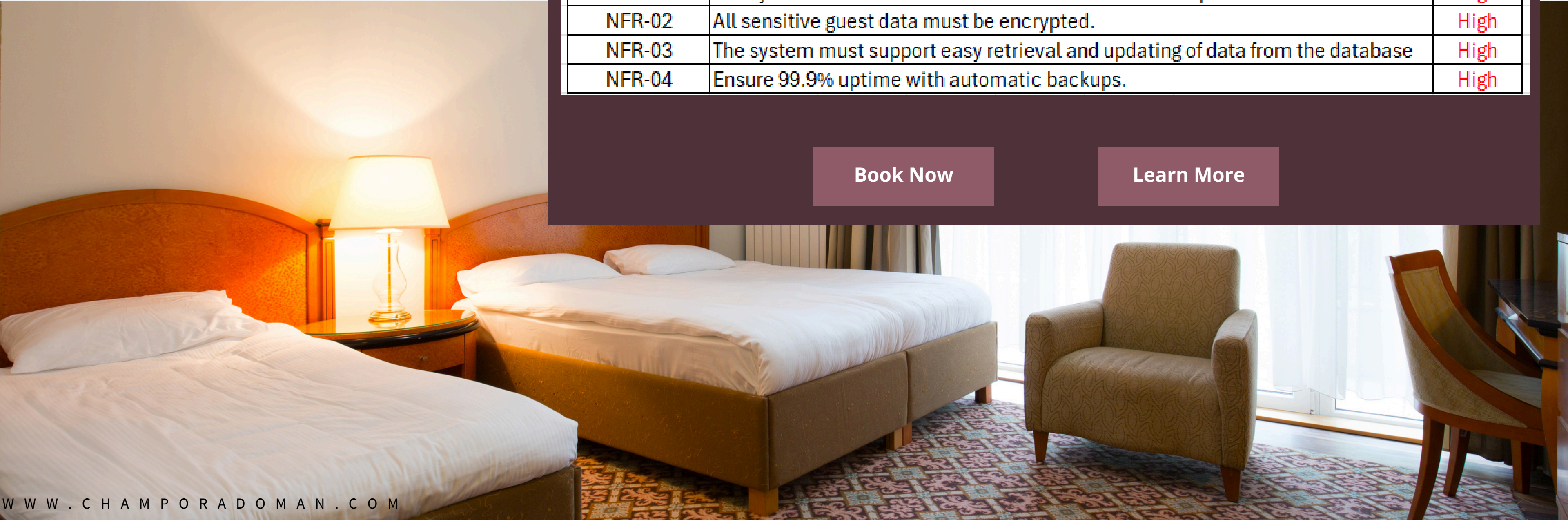


Specific Requirements

Requirement ID	Description	Priority
FR-01	The system must allow online and in-person room reservations.	High
FR-02	Room availability must be updated in real time after each booking.	High
FR-03	System must generate bills for guests at checkout, including additional services	Medium
FR-04	The system keeps track of the amount of inventories used.	High
FR-05	Reports on occupancy and revenue must be available to the manager.	High
NFR-01	The system must handle 100 concurrent users without performance issues.	High
NFR-02	All sensitive guest data must be encrypted.	High
NFR-03	The system must support easy retrieval and updating of data from the database	High
NFR-04	Ensure 99.9% uptime with automatic backups.	High

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Functional Requirements

Room Reservation

Reservations of rooms can be done conveniently on the site or at the hotel, with the availability of the rooms being automatically updated at all times.

Check-in/Check-out

Simplification of the processes of check-in and check-out of the guests is performed by the system, which brings the state of the occupied room to the “occupied” or “available” as it's checked out and finalized.

Billing

The HMS combines all amounts of money which were spent by the guest during his or her stay in the hotel and allows the payment for the services provided via a cash register (credit card, etc).

Inventory Management

The software provides a record of the rooms’ supplies utilization and informs management when the stock of the given items is finished.

Reporting

HMS provides almost all the reports to the hotel operators, which include – volume of guests served, monetary turnover, number of sold rooms and many other statistical indicators of the hotel performance. Times for Reports include Daily, Weekly, and Monthly.



Non-Functional Requirements



Performance

- The system should support up to 100 users connecting concurrently, the response to the request should be delivered within 2 seconds.
- Must handle bookings and check in and out over 500+ rooms.



Security

All the most sensitive data collected from the guests like their details and even credit card information will be encrypted regardless whether it is stored or in transit for example through the use of SSL or TLS.



Usability

- The system must be user friendly to the users for easy access to retrieval and updating of data from the database like the booking, check in / out and billing.
- The system should be able to operate effectively in a web based environment on a computer and also on a mobile phone allowing staff to access data easily for operations



Reliability

The system availability should be at least 99.9 % that entails better performance of continuous operations in the system.



User Needs

Hotel Staff Needs

- Efficient Reservation Management
- Streamlined Check-in/Check-out Process
- Billing and Payment Options
- Inventory Tracking
- Easy-to-Use Reporting Tools

Guest Needs

- Simple and Quick Booking Experience
- Fast Check-in and Check-out
- Secure Payment Options
- Reliable and Secure System

Management Needs

- Centralized Control Over Hotel Operations
- Data-Driven Insights
- Scalability and Reliability





System Features

Feature	Description	Acceptance Criteria
Room Booking and Reservation	Guests can make online and offline reservations.	Booking confirmations sent instantly.
Check-in and Check-out	Hotel staff can manage guest check-ins and check-outs.	Only confirmed bookings can proceed to check-in.
Billing and Payment	Automatic bill calculation at check-out.	Final bill must include all room rates and extra services (e.g., dining).
Reporting	Generates reports for hotel managers showing occupancy, revenue, and inventory.	Reports must be available within 30 seconds of the request.



Acceptance Criteria

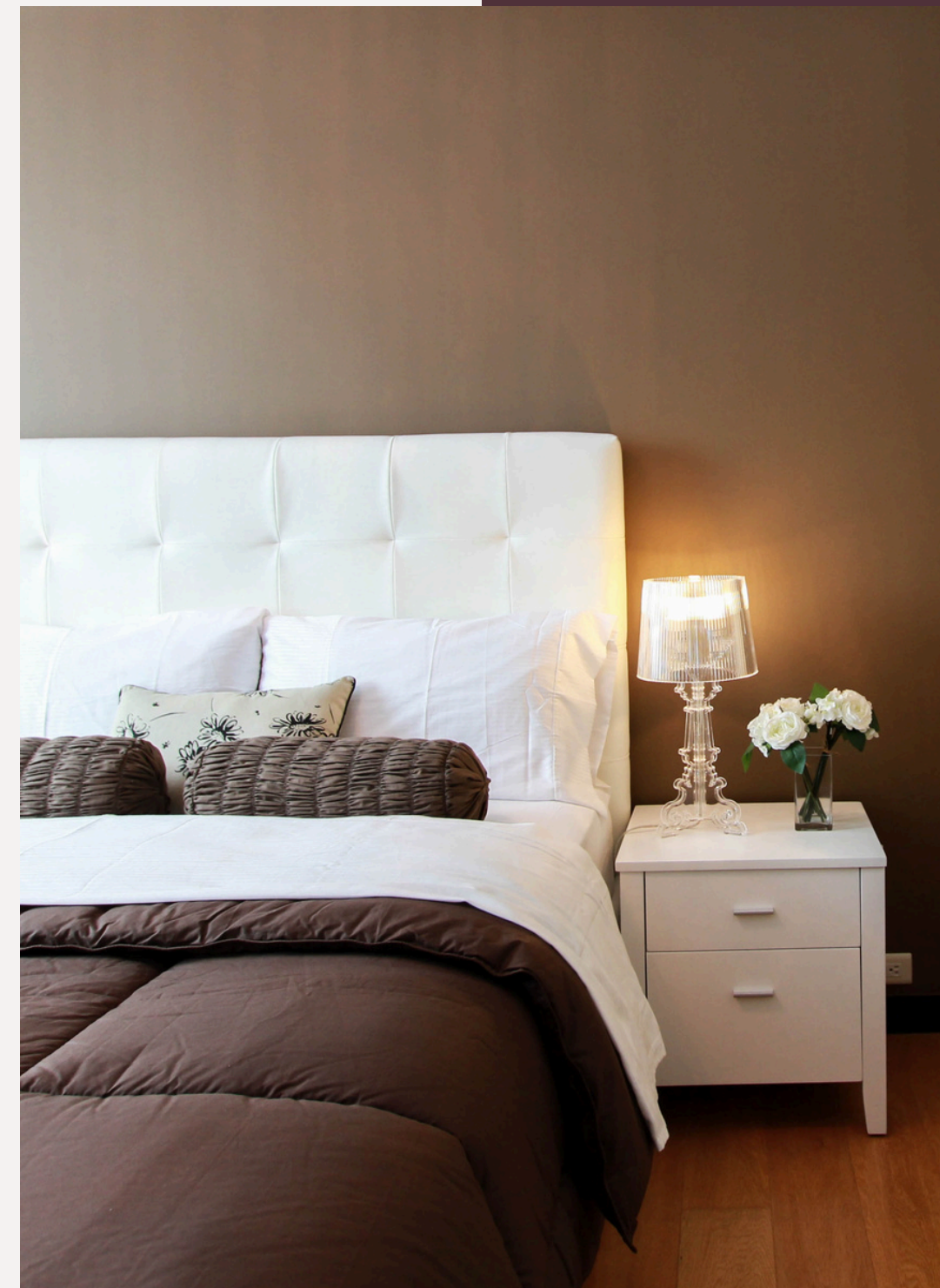
The requirements of both functional and non-functional types need to pass through UAT.

For system acceptance:

All the functionality is tested and is in compliance with the functional requirements.

The non-functional requirements (performance, security, etc.) pass/complete stress and security tests.

The actual users (hotel staff, managers, etc.) approve the usability and performance during the UAT phase.





Appendices

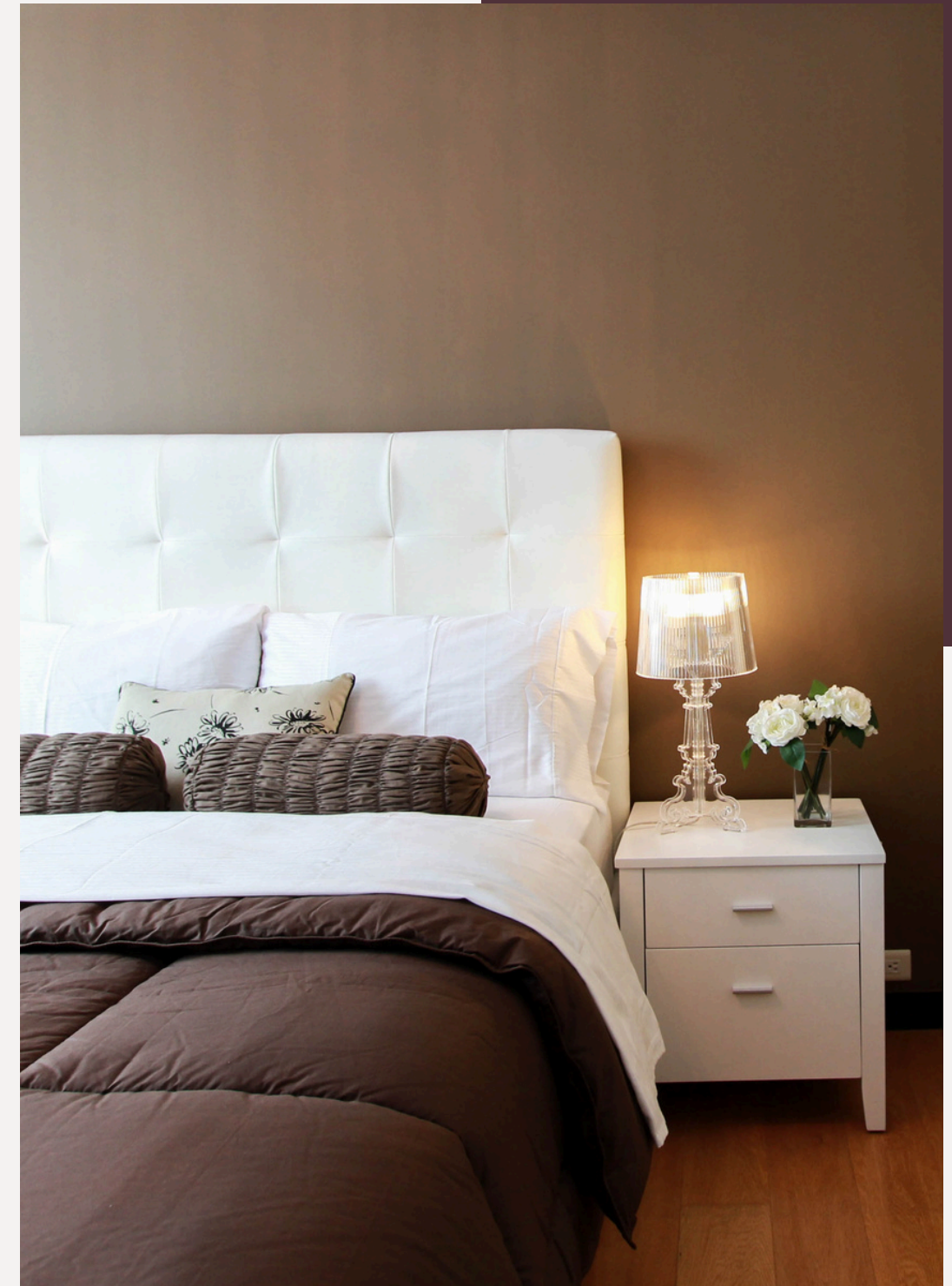
Glossary of Terms:

- HMS (Hotel Management System): provide tailored solutions to keep your hotel business operating optimally.
- OTA (Online Travel Agencies): is an online or app-based marketplace where consumers can browse, book, and pay for hotel rooms, transportation, lodging, airfare, restaurants, and experiences.
- API (Application Programming Interface): is a collection of communication protocols and subroutines used by various programs to communicate between them.
- DB (Database): is an organized collection of data stored in a computer system and usually controlled by a database management system (DBMS).
- POS (Point of Sale): It is a computer-based cash register with software capable of tallying up orders, taking payments, monitoring inventory and buying trends, creating invoices, and collecting marketing data.
- SSL (Secure Socket Layers): is a security protocol that provides privacy, authentication, and integrity to Internet communications. SSL eventually evolved into Transport Layer Security (TLS).
- TLS (Transport Layer Security): is a security protocol that provides privacy and data integrity for Internet communications. Implementing TLS is a standard practice for building secure web apps.
- JSON (JavaScript Object Notation): is a standardized format for representing structured data.
- XML (Extensible Markup Language): is a software- and hardware-independent tool for storing and transporting data.
- UAT (User Acceptance Testing): is the final testing stage in software development before production. It's used to get feedback from users who test the software and its user interface (UI).





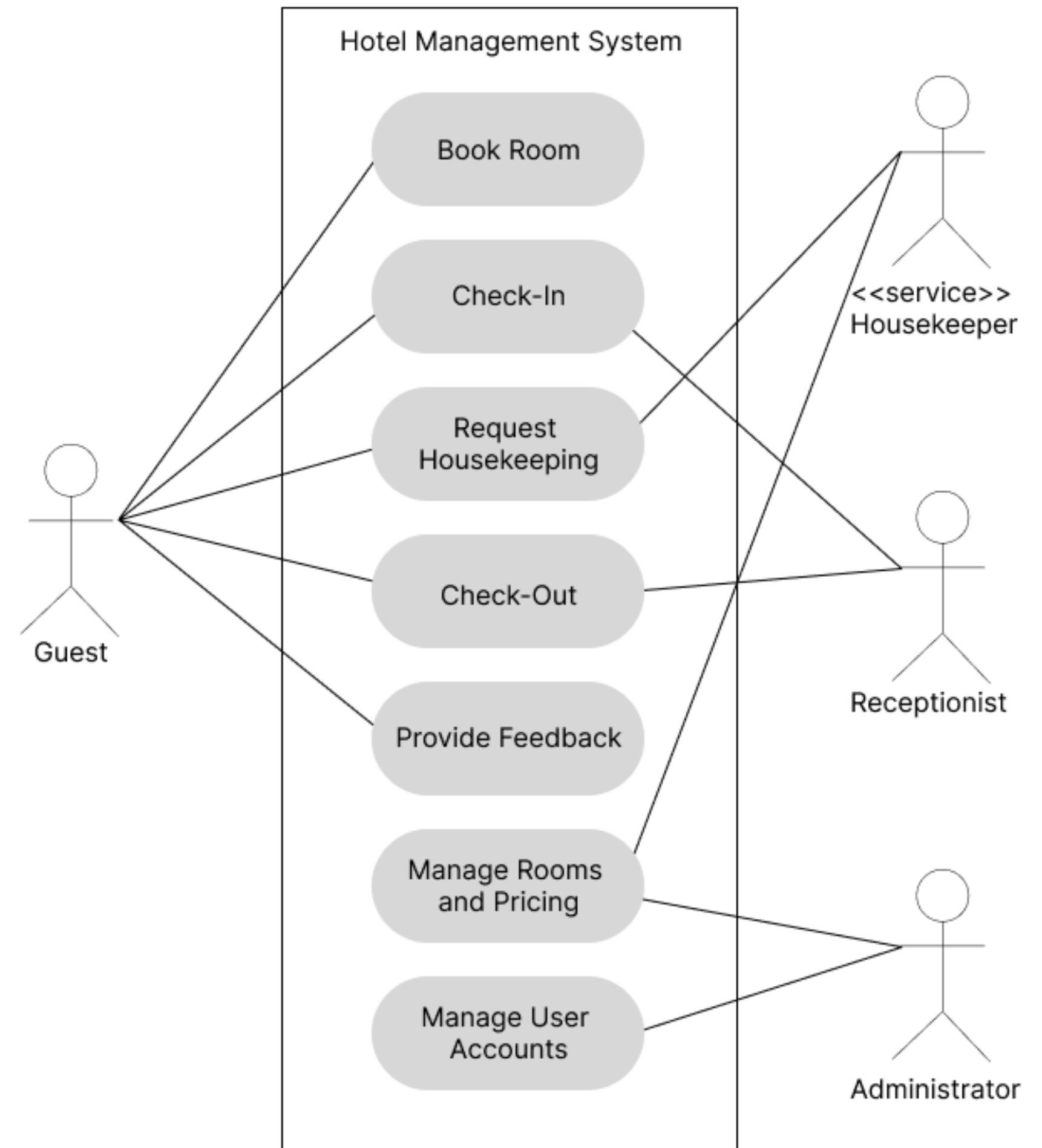
System Diagrams



Case Diagram

USE CASES

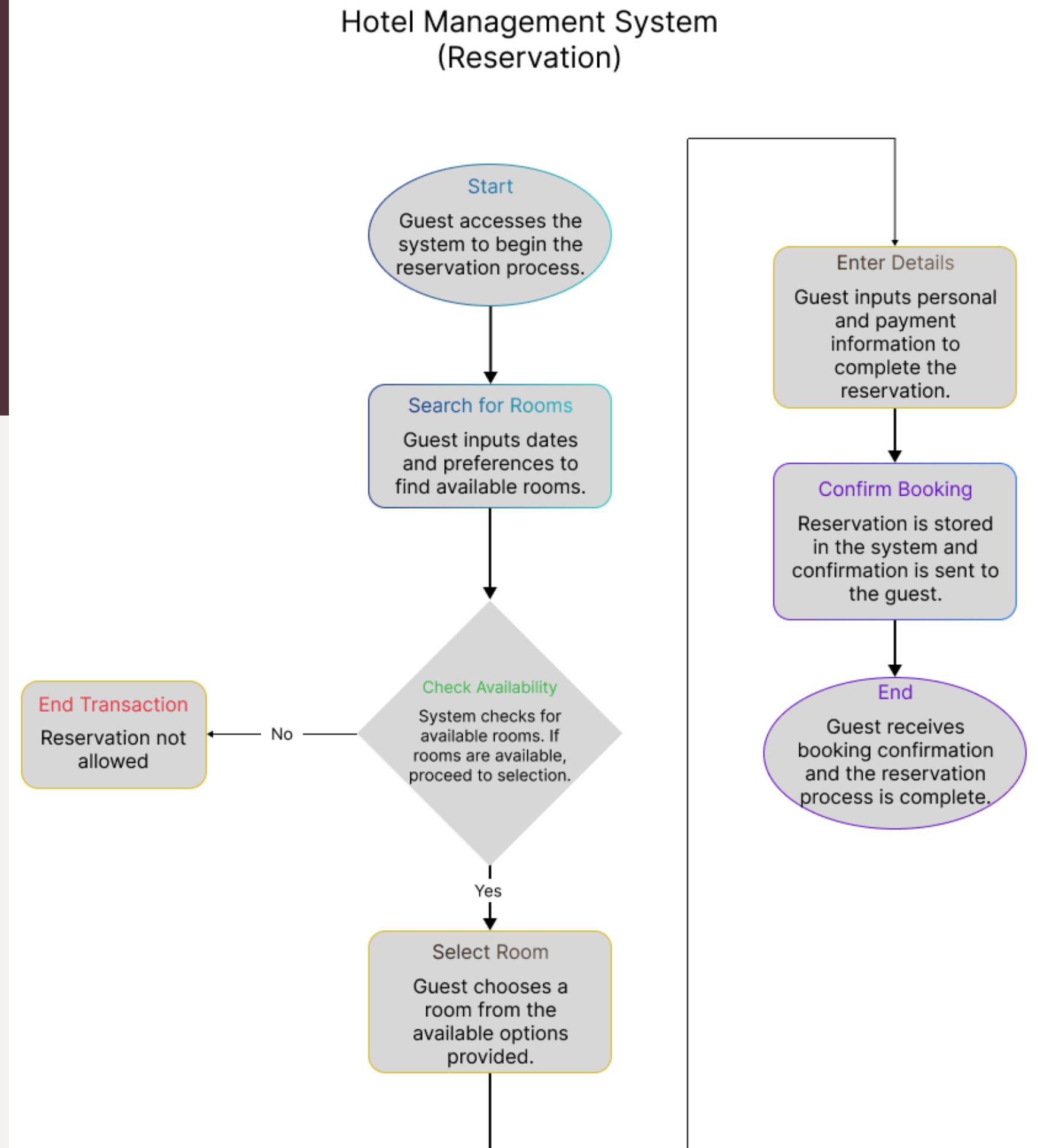
- **Book Room:** Guests can look and make reservations for rooms that they desire. Receptionists can do manual bookings for walk in customers.
- **Check-In:** Guests are given keys and recorded as checked in by the receptionist.
- **Check-Out:** The receptionist changes the room status after the payment process.
- **Request Housekeeping:** Hotel guests can request such services as cleaning and more amenities.
- **Update Room Status:** Housekeeping staff determine the state of a room as either clean or in need of repair.
- **Provide Feedback:** Guests can write comments or feedback based on the hotel's services.
- **Manage Rooms and Pricing:** The room catalogue, the set prices and promos are controlled by the administrator.
- **Manage User Accounts:** The administrators assign staff roles and who can see what services and secure the system.



Activity Diagram

PROCEDURE OF MAKING A RESERVATION

- **Start:** The system is accessed by the guest.
- **Look for available Rooms:** The guest gives specific details such as the dates and preferences for a certain room.
- **Checking the Availability of a Room:** The system provides an answer about the rooms available.
 - Proceed with the process if there are any rooms that are available.
 - End the process if there are no rooms available.
- **Choose a Room:** The room is selected by the guest.
- **Input the Information:** The guest fills personal and billing details.
- **Making the Reservation:** The system logs and reserves the details and sends a message with confirmation of booking.
- **End:** The guest gets a message with confirmation of the initiated booking ending the process.



Class Diagram

Hotel

Attributes
hotelName , hotelAddress , contactInfo , generateReport

Relationship
One hotel can have multiple **Staff** and multiple **Rooms**.
A hotel can generate 0 or more reports

Booking

Attributes
bookingID , checkInDate , checkOutDate , #guest , #room , totalPrice , bookingStatus

Relationship
A booking is associated with one guest and one room. A room can have multiple bookings over time.

FeedBack

Attributes
feedbackID, guestID , feedbackText , feedbackDate

Relationship
A guest can make zero to many feedback

Room

Attributes
roomNumber , roomType , roomStatus , roomPrice

Relationship
One hotel can have many rooms, and each room can have one **Booking**.

Guest

Attributes
guestID , guestName , contactDetails , bookingHistory

Relationship
A guest can make multiple bookings, but a booking is associated with only one guest.

Amenity

Attributes
amenityID, amenityName , amenityDescription , amenityPrice

Relationship
A room can have zero to many amenities.

Report

Attributes
reportID , reportType , generateDate

Relationship
Each report is belongs to the specific hotel

Staff

Attributes
staffID , staffName , staffRole , contactInfo

Relationship
A hotel can have multiple staff members.

Payment

Attributes
paymentID , paymentAmount , paymentMethod

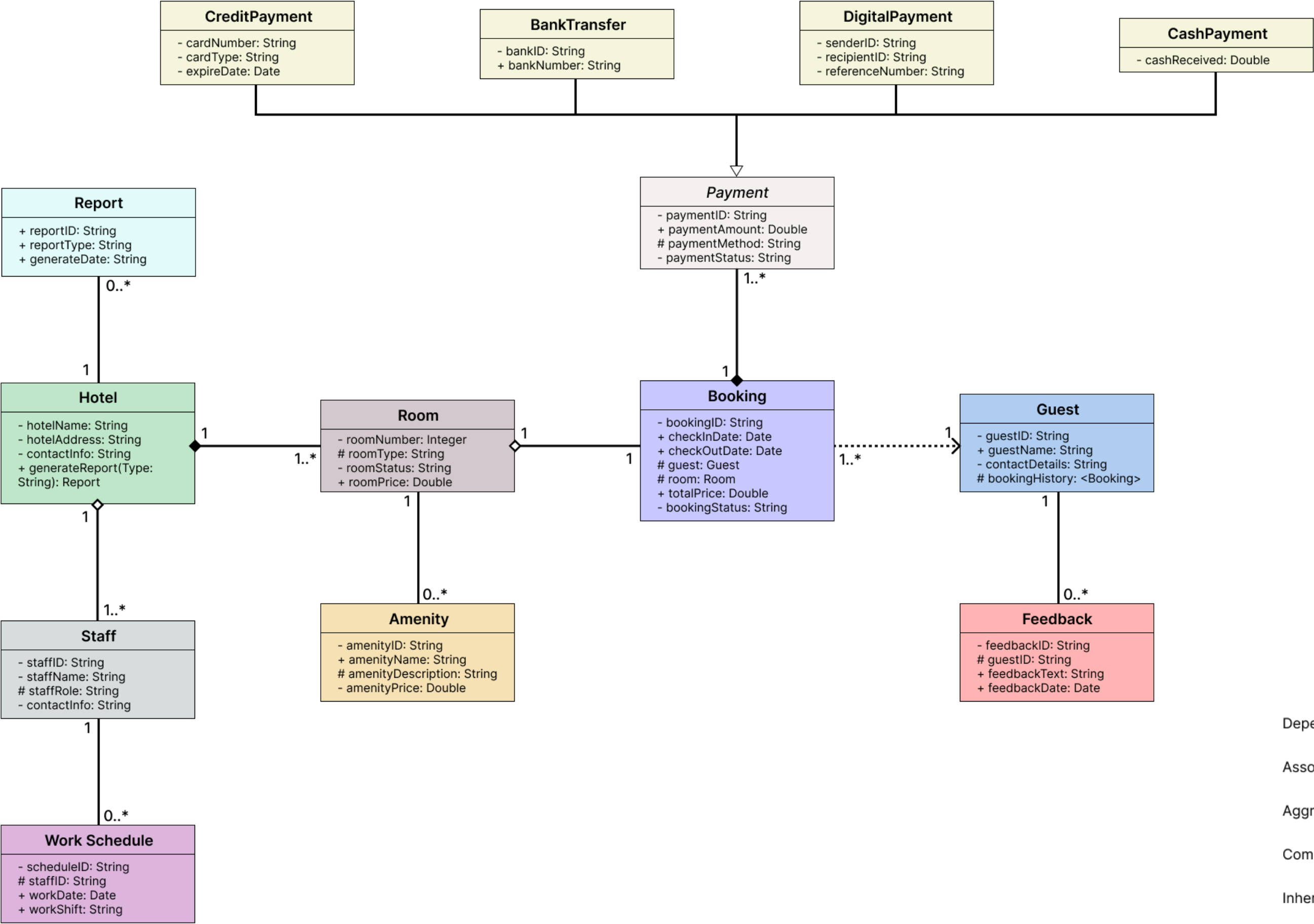
Relationship
A booking can have multiple payments.

Work Schedule

Attributes
scheduleID, staffID , workDate , workShift

Relationship
A staff can have zero to many work schedule depending on the job

Class Diagram



Package Diagram

TOP-LEVEL PACKAGE: HOTEL MANAGEMENT

This section has procedures outlines for all levels; delegation of responsibilities, reporting and, analytics.

SUB-PACKAGES

Staff: Assign in relation to scheduling, task distribution and subordinate performance.

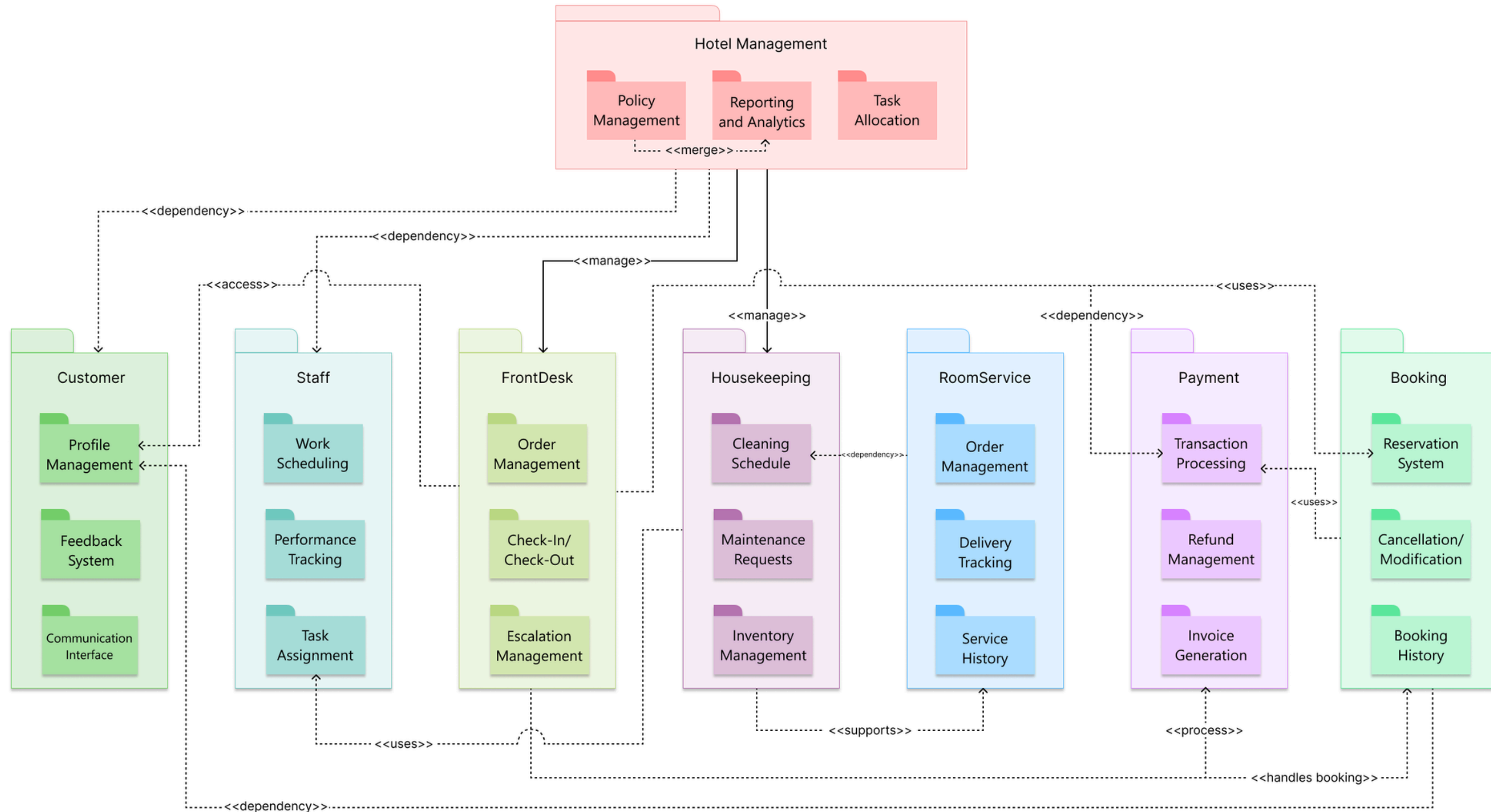
FrontDesk: Interact with the customer during the registration, clients' queries processes etc.

Customer: Manage the customer profile and feedback.

RoomService and Housekeeping: In charge for preparation of the room for room service and its delivery.

Booking and Payment: Deals with booking activities and the payment activities.

Package Diagram



Sequence Diagram

CUSTOMER INTERACTION

- The process is initiated when the customer wishes to locate an empty room.
- Once the room has been confirmed as available, its description is provided to the customer.
- The customer decides upon a room and begins the booking process.

HOTEL SYSTEM OPERATIONS

- When there are lots of free rooms, all of them are enhanced by detailed room descriptions stored in the system and these are displayed to the customers.
- If a room has already been selected, the preferred occupation is booked, and payment is made.

PAYMENT PROCESS

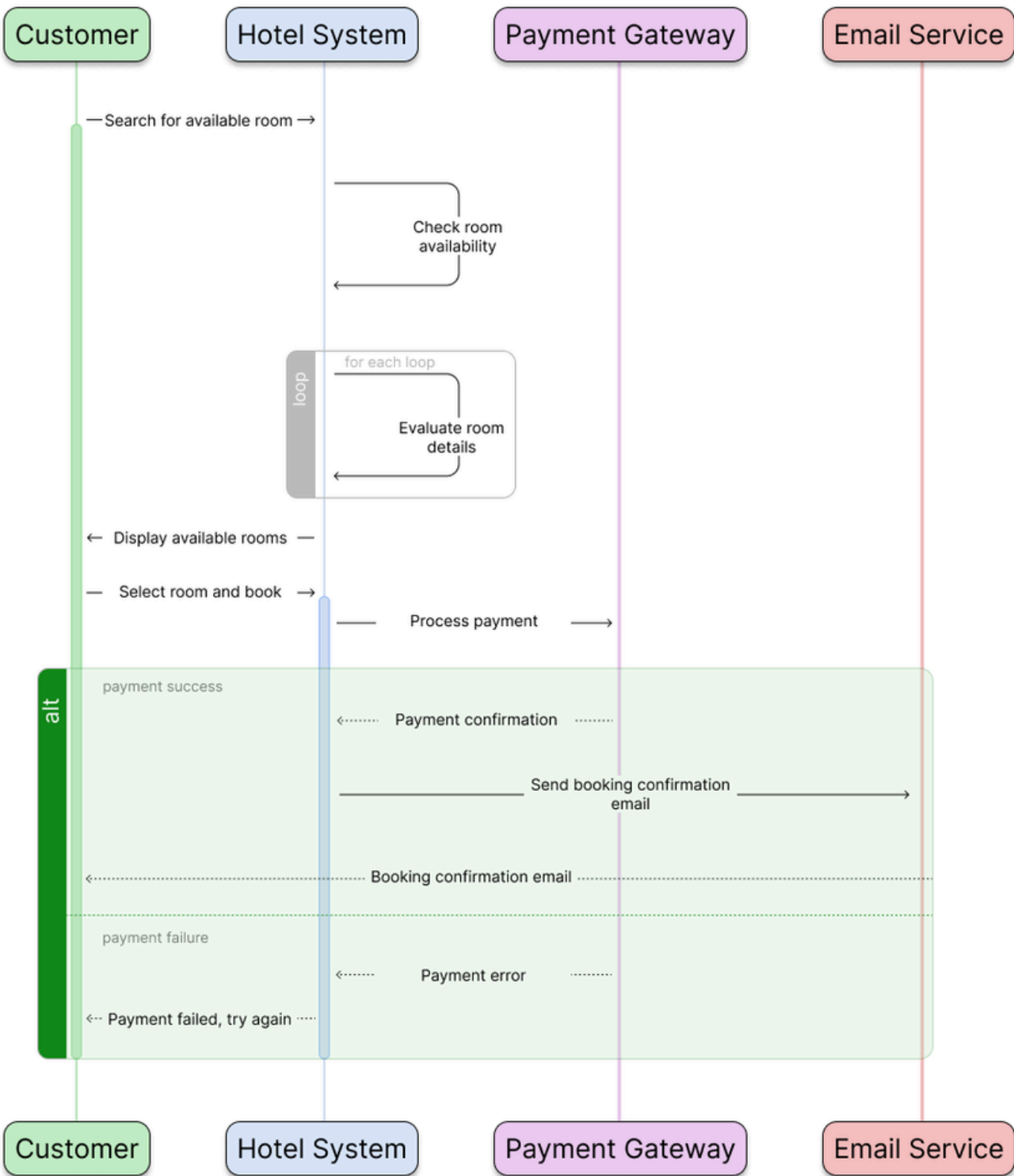
The request for payment is generated by the system for the Payment Gateway.
The possible possibilities are:

- **Payment Processed:** The Payment Gateway processes the payment, and the Hotel System automatic booking confirmation email has been sent via Email service. The email then reaches the customer’s inbox finalizing the process.
- **Payment Not Processed:** In case the payment has not gone through, the customer would be highlighted by the system which would drive them to attempt for the second time.

EMAIL SERVICE

When the booking has been made and the payment has been completed, the Email Service will respond to the customer with an email confirming their reservation.

Hotel Management System - Room Booking Process



Data Flow Diagram

DIAGRAM FLOW

IDENTIFY ROOM

The first step in the procedure is where the user selects a room which they would like to book.

The room ID selected is picked by the system for use.

LOOK UP FOR AVAILABLE ROOM

To find the room chosen by the user and its cost, the system goes to the RoomPrice data base.

The searches of the room and price availabilities are done, and results are moved to the next step.

COMPUTE TOTAL COST OF ROOM

The system can determine the total amount by the price of the room. Determines and gets ready the total cost for payment.

SETTLE TRANSACTION AND ISSUE RECEIPT

The last step is completed: the client paid the full amount for all transactions. The receipt is provided to the customer immediately after the payment has occurred.

KEY ELEMENTS IN THE DIAGRAM

ENTITIES

Customer: This is the outside actor which triggers the system.

PROCESSES

These are actions taken in the system. To include Identify Room, Look up for Available Room, Compute Total Cost of Room, Settle Transaction and issue receipt.

DATA STORE

RoomPrice: Contains the information pertaining to the rooms and their prices.

DATA FLOW

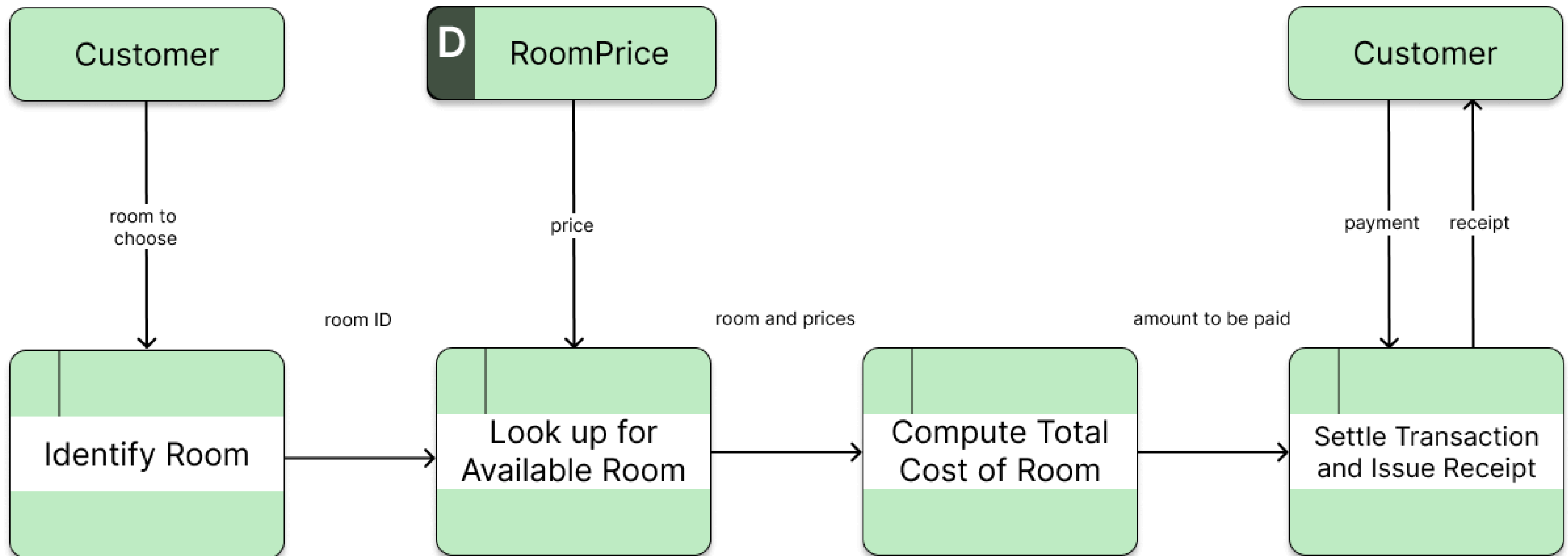
The arrows show what is moving in between entities, processes and datastores. For instance:

room to choose moves from a customer to an Identify Room process.

“Price” is acquired from the RoomPrice database.

Data Flow Diagram

Logical Data Flow Diagram - Hotel Booking System





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Thank You

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